belinus®

Powerbox · S1-5
v01 · 2026-01
MODEL · HWRIS1-5-001

RESIDENTIAL HYBRID INVERTER · SIC · SINGLE-PHASE 5 KW

Powerbox S1-5

Unlock the power of nature.

#ResidentialHybridInverter #SiC #98.2%MaxEfficiency #SinglePhase #5kW #2BatteryPorts

01 **SiC technology.** Silicon-Carbide based — significant performance advantages and lower cost of ownership.

02 **3 MPPTs of 20 A.** 150 % to 200 % input power with AFCI / PID function.

03 **Ultra-thin design.** 165 mm deep, 36 kg, IP65 — indoor or outdoor installation.

04 **150–340 % grid input power.** Max. 18 kVA, parallel load and charging. Up to 400 % overload.

05 **2 independent battery ports.** Low-voltage 30 – 60 V, 250 A total charge / discharge.

06 **Integrated belinus EMS.** Energy Management System built in.

10 yr	10 yr	10 yr
PRODUCT WARRANTY	PERFORMANCE WARRANTY	SERVICE WARRANTY

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TECHNICAL SPECIFICATIONS

Powerbox S1-5 · Single-phase Hybrid Inverter

RATED OUTPUT	MAX EFFICIENCY	MPPT TRACKERS	BATTERY PORTS
5 kW	98.2 %	3	2

GENERAL

Weight	36 kg
Dimensions (W × D × H)	465 × 165 × 665 mm
Noise	< 45 dB(A)
Operating temperature	−25 °C to 60 °C
Cooling	Intelligent air cooling
Ingress protection	IP65
Altitude	4000 m

OFF-GRID OUTPUT

Rated power	5 kW
Max. power	5.5 kVA
Peak power	20 kVA (10 s)
Max. current	25 A
Output voltage	220 / 230 / 240 V
Switching time	< 10 ms

PV INPUT

Max. input power	18 kW
MPPT voltage range	130 – 800 V
Start voltage	140 V
Max. voltage	800 V
Number of MPPT	3
Max. current	20 A × 3

ON-GRID

Max. input power	18 kVA
Grid frequency	50 – 60 Hz (±5 Hz)

BATTERY

Battery type	belinus Energywall
Inputs	2
Voltage range	30 – 60 V
Charge / discharge	125 A × 2

EFFICIENCY

Max. efficiency	98.2 %
Battery charge / discharge	95.2 %

CERTIFICATION

CE-LVD	IEC 62109-1&2 · 62477-1
Grid	VDE-AR-N 4105 · EN 50549-1 · AS/NZS 4777.2
Monitoring	LED · WiFi · 4G · Bluetooth · RS485

PROTECTION

Functions	Anti-islanding · RCM · AFCI
Surge	DC Type II / AC Type III
Comms ports	RS485 · CAN · DRED · Dry contact

Notice: specifications are subject to change without notice.